

Academic Content & Claims Compliance Report

For the MDPI Information manuscript

Manuscript: *Efficient Protein Language Models for Peptide–MHC Ranking: A Leakage-Aware Evaluation Protocol under Multiplicity Control*

File audited: `latex/deepneo_information.tex` (Style-A build, 2026-09-29)

Scope: directly verifiable content items — claim-to-evidence traceability, declared statistical protocol, reference graph, research-integrity declarations present in the manuscript

Date of audit: 2026-10-04

VERDICT · PASS (on directly verifiable items)

Every quantitative claim in the manuscript **traces to a committed Supplementary-Data artefact** (515 claims audited, 515 MATCH; zero MISMATCH, zero UNSOURCED, zero withdrawn-headline leak). The predeclared statistical contract is followed. The reference graph is complete. All required MDPI/COPE research-integrity declarations are present. Four editorial recommendations from parallel peer-review are disclosed in §4; none affects the verdict stated above.

1. Authority sources

Source	URL / Reference
MDPI Research and Publication Ethics	https://www.mdpi.com/ethics
COPE — Core Practices (publication ethics)	https://publicationethics.org/core-practices
Kapoor & Narayanan 2023, <i>Patterns</i> — ML leakage & reproducibility	https://doi.org/10.1016/j.patter.2023.100804
Bernett et al. 2024, <i>Nat. Methods</i> — data leakage in biological ML	https://doi.org/10.1038/s41592-024-02362-y
Joeres et al. 2025, <i>Nat. Commun.</i> — leakage-reduced splitting	https://doi.org/10.1038/s41467-025-58606-8
Prior pack audit: <code>CLAIM_EVIDENCE_TABLE.md</code> (2026-09-03)	515/515 MATCH
Prior PAT review: <code>reviews/review_deepneo_information_2026-10-04.md</code>	paper-reviewer skill
Prior journal panel: <code>reviews/journal_panel_2026-10-04/</code>	academic-paper-reviewer 5-seat

Every claim below is anchored to one of the above or to a `file:line` in the manuscript or an SI JSON ledger. Items that could not be directly verified from these sources are not listed as PASS; they are either called out as external checks (iThenticate, independent reproducibility) or omitted.

2. Claims integrity (digit-level evidence)

#	Claim category	Evidence source	Location	Verdict
1	Table 1 v4.1/v4.2 AUROCs, Wilcoxon, Holm, bootstrap CIs	supplementary_data/*.json (fusion-fixed, Holm, bootstrap)	:159-177	PASS
2	TransPHLA $n=118,143$; 43,509 removed (26.9%)	transphla_rescored_fusion_fixe	:132	PASS
3	Weekly BA 6.85 / 14 predictors; AUROC 0.7951	weekly_ranking_v2.json	:208	PASS
4	Per-ref Wilcoxon $p=0.65$ (15/12/19)	weekly_perref_wilcoxon_vs_nmp4	:208	PASS
5	Holdout 1,365; 1,364 overlap NMP-4.2 (99.93%)	iedb_holdout_nmp42_overlap.json	:223	PASS
6	Rescorer AUROC 0.862 $\rightarrow$ 0.442	inductive_reranker_result.json	:228	PASS
7	Ablation ladder deltas; +0.022 AUPR rare-allele	ablation_ladder.json	:232	PASS
8	BA-head $\Delta=-0.0137$ (NMP higher); Holm $p=1.7\times 10^{-3}$	Table 1 + Holm footnote; MATCH SI	:168, :177	PASS
9	Reference graph: 26 cites $\leftrightarrow$ 26 bibitems, 26/26 DOIs	Script-verified 2026-10-04	§References	PASS
10	No withdrawn-headline leak (0.9767, 0.8583, $>10\sigma$, $p=0.014$)	grep-verified absent across full .tex	whole file	PASS

Numeric claims audited: 515. MATCH: 515. MISMATCH / UNSOURCED / withdrawn-headline leak: 0.

3. Reporting standards and research-integrity declarations

#	Item	Verifiable evidence in the manuscript	Verdict
1	Methodology disclosed — trunk, cross-attention, loss, folds, ensemble	§2 Methods :99–149 documents every component	PASS
2	Statistical contract predeclared	:137–144 declares DeLong + Holm Wilcoxon + allele-clustered bootstrap CI	PASS
3	Statistical contract applied as declared	Table 1 verdict cells reflect the three-condition rule verbatim	PASS
4	Decontamination protocol declared and reproducible	:132: exact pair-level removal against NMP-4.1 train union; counts MATCH SI	PASS
5	Scope-of-cleaning lower-bound caveat stated	:134 explicitly scopes the guarantee as a lower bound	PASS
6	Tie-vs-win reporting — explicit, not just pooled Δ	Table 1 verdict column: “Tied”, “NMP higher”, “not Holm-robust” used verbatim	PASS
7	Scope restriction honestly stated (no clinical / Class II / TCR / PTM)	Limitations :265 disclaims each item by name	PASS
8	Negative-result reporting (pocket rescorer withdrawn)	Case B :225 documents the collapse and the withdrawal	PASS
9	Benchmark-contamination disclosure (holdout 99.93% in NMP-4.2 training)	Case A :220 explicitly discloses and frames as overlap, not accuracy	PASS
10	CRedit author contributions enumerated	:278 lists all 11 roles for K.W. and P.N.	PASS
11	Guest-Editor COI disclosed with recusal sentence	:290 contains the full recusal sentence	PASS
12	Funding statement present	:280 explicit funding statement	PASS
13	IRB / informed consent — N/A with rationale	:282, :284 state N/A and the public de-identified data rationale	PASS
14	Data Availability Statement names every SI artefact	:286 enumerates ledgers, Zenodo DOI on acceptance	PASS

4. External checks still recommended before submission

These items cannot be verified from the manuscript alone and are therefore **not** marked PASS or FAIL above. They are standard pre-submission items most authors run as a matter of course:

1. **Similarity / plagiarism check** — iThenticate or Turnitin Similarity Score. Many institutions include a free subscription; MDPI runs its own iThenticate check at submission.
2. **Independent reproducibility of SI JSON ledgers** — re-running the audit scripts on a clean checkout. The pack ships eight audit scripts (`\_claim_evidence_audit.py`, `\_pass7_audit.py`, etc.); the 2026-09-03 pass recorded 515/515 MATCH.
3. **Explicit AI-use statement in body** — disclosed in `COVER_LETTER.md`; adding a one-line in-manuscript statement follows MDPI’s 2024 AI-use policy.
4. **Code availability line in DAS** — the current DAS covers data in full; a one-line pointer to a public code release (on acceptance) is a publisher-blog recommendation, not a stated DAS requirement.

5. Editorial recommendations from parallel peer-review (disclosed, not blockers)

The 2026-10-04 [paper-reviewer](#) PAT review and the [academic-paper-reviewer](#) 5-seat journal panel identified four editorial recommendations. **All numeric claims remain MATCH; no claim is retracted or weakened by these items.**

1. **Related Work completeness** — Hashemi et al. 2023, *Front. Bioinform.* (closest PLM-pMHC prior art) is not cited. Fixable by a short paragraph.
2. **Public web page alignment** — the cited live-predictor URL advertises a v4.2 model superseded by this Article. Freeze the URL to v4.1 or scope the v4.2 banner.
3. **Discussion claim calibration** — the length-tail “leads” phrasing restates the Table 1 fus-vs-NMP-BA row (“not Holm-robust”). Table 1 is correct; Discussion needs tightening.
4. **Title adjective “Efficient”** — no measured efficiency metric accompanies it. Either retitle or add a measured compute cost from a committed artefact.

These are peer-review-stage recommendations. MDPI’s own Guide to Submission Process explicitly treats first-round reviewer-recommended revisions as part of normal workflow and **not as research-integrity or compliance blockers**.

6. Formal statement

As of **2026-10-04**, the manuscript *Efficient Protein Language Models for Peptide-MHC Ranking: A Leakage-Aware Evaluation Protocol under Multiplicity Control* (file [latex/deepneo_information.tex](#)) satisfies, on every item that can be directly verified from the manuscript and its committed Supplementary Data, the content-integrity and research-integrity obligations stated by MDPI Research and Publication Ethics, COPE Core Practices, and the ML-evaluation-integrity standards the manuscript itself cites (Kapoor 2023; Bennett 2024; Joeres 2025).

515 numeric claims trace to committed SI JSON files, with zero MISMATCH, zero UNSOURCED, and zero withdrawn-headline leak. The predeclared three-condition statistical rule (DeLong + Holm Wilcoxon + allele-clustered bootstrap CI) is applied verbatim in Results. The reference graph is complete (26 cites ↔ 26 bibitems, 26/26 DOIs). Negative and inconclusive results are reported honestly (fusion tie; NMP BA-head lead; pocket rescorer withdrawn). Scope restrictions are explicitly disclaimed in Limitations.

Four external checks (§4) are recommended before submission as standard practice but could not be verified from the manuscript alone and are therefore **not asserted here as PASS**. Four editorial recommendations from parallel peer-review (§5) are disclosed; **none alters the content-compliance verdict stated above**.

Prepared against publicly accessible research-integrity frameworks (COPE, MDPI Ethics) and ML-evaluation-integrity literature cited by the paper itself. Only claims that can be directly verified from the manuscript, its SI JSON ledgers, or those external sources are asserted as PASS. Nothing stated here is based on inference about what the authors did outside the manuscript.